Sequence 6

FUNCTIONS: REPRESENTATIONS AND REFERENCE FUNCTIONS

23 November–4 December 2026

I Function, image and preimage

Definition

A real-valued function associates to each real number x of its domain D_f a unique real number $f(x)$, called the image of x. A preimage of y is a real number $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no preimage, one, or several.

Example 6.1

Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; determine the preimage of 6 under h.

Solution:

Example 6.2

For the square function, calculate the images of -3 and 3, then the preimages of 9, 0 and -1.

Solution:

```
def g(x):
    return x*x - 4*x + 32
```

Example 6.3

Distinguish `g(-5)` from `x = -5`.

Solution:

II Domain of definition

Conditions for existence

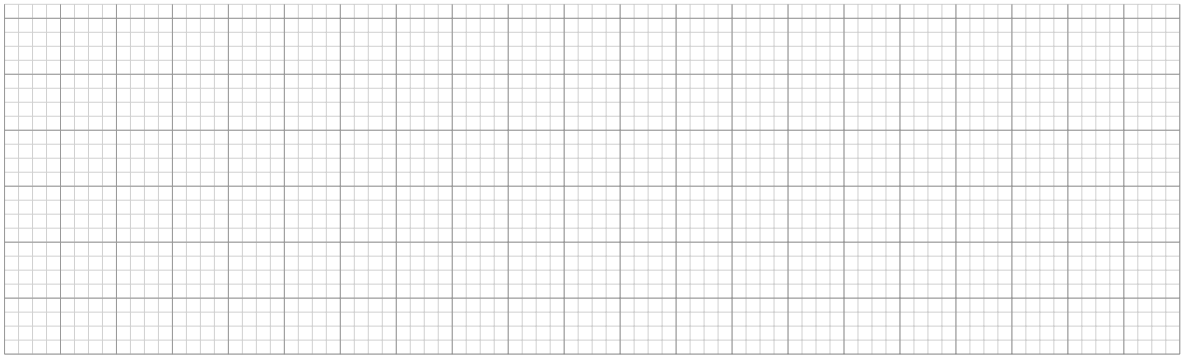
D_f contains the values for which the process defining f makes sense. A denominator must be nonzero; a number under a square root must be positive or zero. Also respect any domain imposed by the statement or the context.

Example 6.4

Determine the domains of definition:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$

Solution:

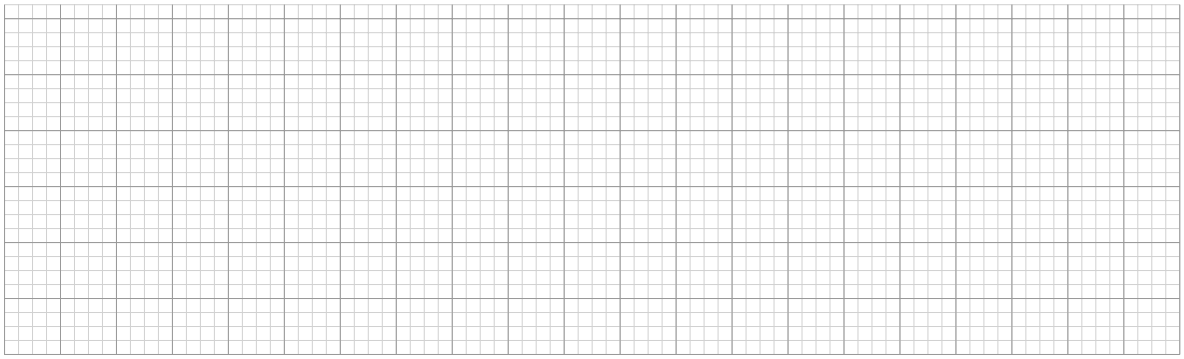


Watch out. The square root of a negative number does not exist in $\mathbb{R}$; it is not "a negative square root" that is excluded.

Example 6.5: Modelling

A rectangle of perimeter 20 m has width x m. Express its area and state the domain if both dimensions must be strictly positive.

Solution:



III Table of values and representative curve

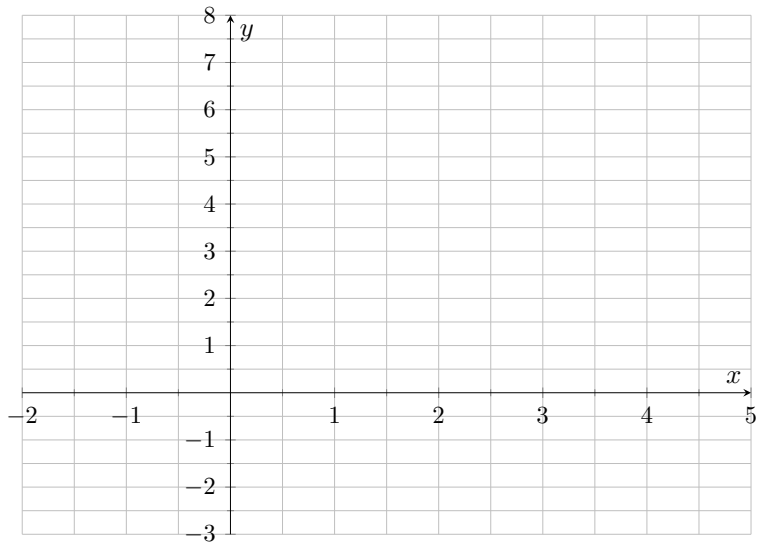
Definition

$\mathcal{C}_f$ is the set of points $(x; f(x))$, with $x \in D_f$. Thus $M(a;b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

Example 6.6

On $[-1;4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then draw the curve over the whole interval.

x	-1	-0.5	0	1	1.5	2	3	4
$f(x)$	.....	.....	-1	-2	.....	-1	.....	.....



Solution:

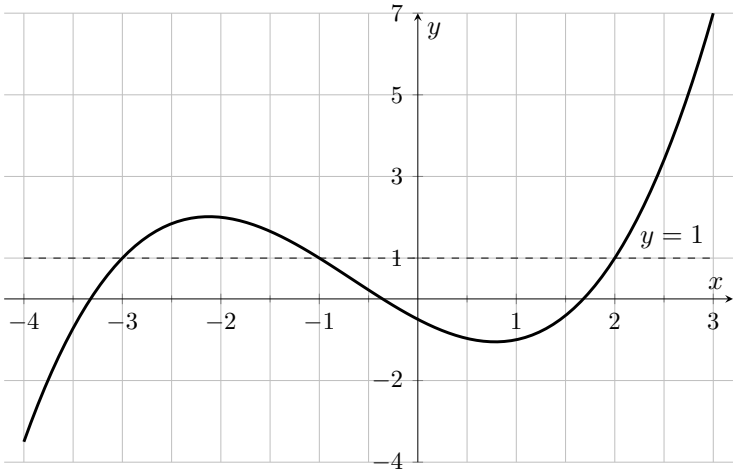
IV Graphical solving of equations and inequalities

Method

The solutions of $f(x) = k$ are the abscissae of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read the abscissae where the curve is below this line, intersections included; with $<$, the intersections are excluded.

Example 6.7

The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4;3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



Solution:

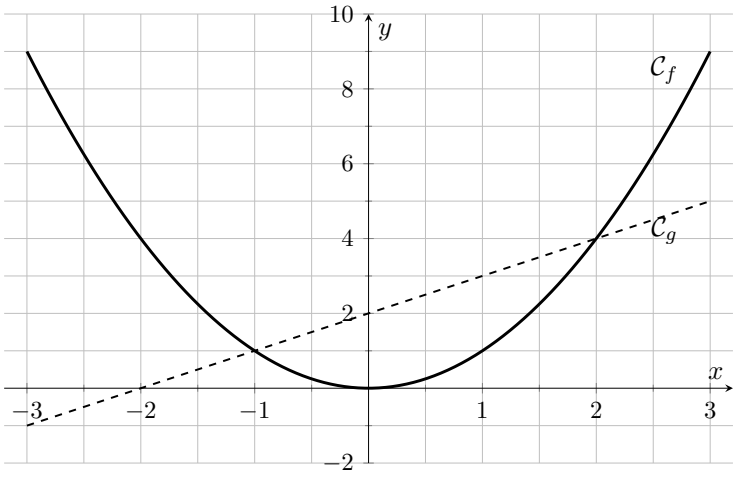
V Comparing two functions; reading a sign

Method

$f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below $\mathcal{C}_g$ at the same abscissa. The sign of $f(x)$ is read relative to the x-axis.

Example 6.8

On $[-3;3]$, $f(x) = x^2$ and $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of g.



Solution:

VI Affine functions and absolute value

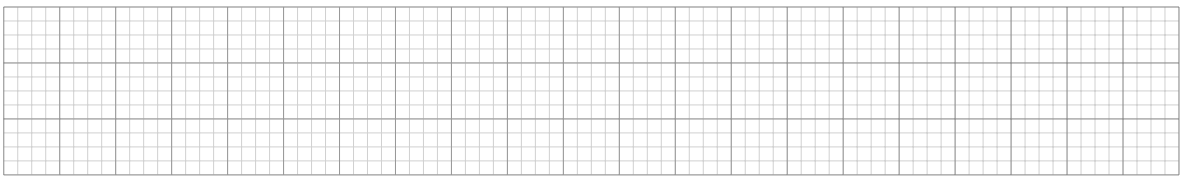
Definitions

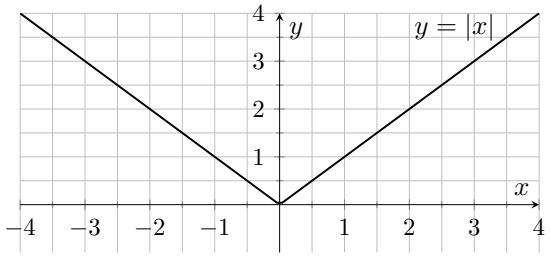
An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of m to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at 0.

Example 6.9

Classify (several answers possible) as affine, linear, constant: $5x + 2$; $5x$; $7 + 2x - 7$; 6; $6(4x - 2)$; $6x + 5 - 6x$.

Solution:





Equations and inequalities

If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$|x| = k \iff x \in \{-k; k\}, \quad |x| \leq k \iff x \in [-k; k],$$

$$|x| \geq k \iff x \in]-\infty; -k] \cup [k; +\infty[.$$

Example 6.10

Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

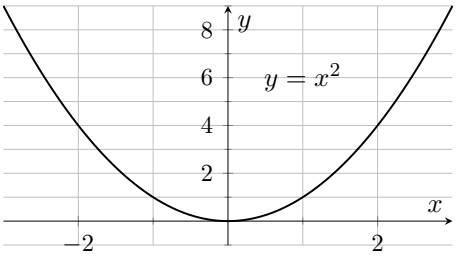
Solution:

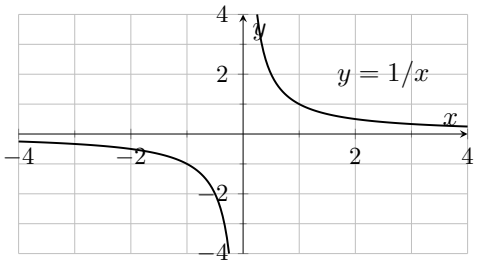


VII Square and inverse functions

Definitions and sign

$x \mapsto x^2$ is defined on $\mathbb{R}$; its curve is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its curve is a hyperbola. $1/x$ has the sign of x and never vanishes.





Equations and inequalities with x^2

If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}; \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}; \sqrt{k}],$$

$$x^2 \geq k \iff x \in]-\infty; -\sqrt{k}] \cup [\sqrt{k}; +\infty[.$$

Example 6.11

Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

Solution:

VIII Inequalities with the inverse function

Method

Respect $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by x without knowing its sign.

	$k < 0$	$k = 0$	$k > 0$
$1/x \leq k$	$[1/k; 0[$	$]-\infty; 0[$	$]-\infty; 0[\cup[1/k; +\infty[$
$1/x \geq k$	$]-\infty; 1/k]\cup]0; +\infty[$	$]0; +\infty[$	$]0; 1/k]$

Example 6.12

Solve $1/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

Solution:

Example 6.13

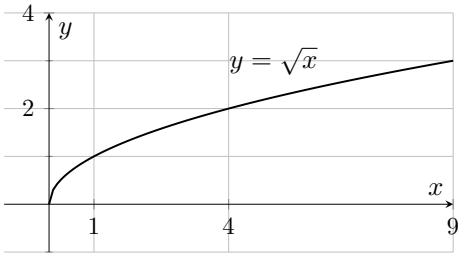
A student writes: "$1/x \leq 2 \iff 1 \leq 2x$, so $x \geq 1/2$". What is missing?

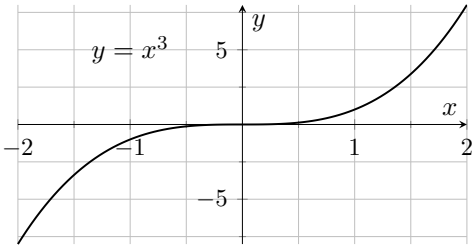
Solution:

IX Using square root and cube

Definitions

$x \mapsto \sqrt{x}$ is defined on $[0; +\infty[$ and takes values that are positive or zero. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Every real number k has a unique preimage, written $\sqrt[3]{k}$.





Example 6.14

Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

Solution:

Example 6.15

Compare x^2 and x for $x = 1/2$, then $x = 2$. Is the square of a positive number always greater than the number itself?

Solution: